WEEK 1: ASSIGNMENT 1

Q1. Data analytics is best defined as:
a) Collection of large volumes of data
b) Storage of structured and unstructured data
c) Scientific process of transforming data into insights for decision-making
d) Visualization of data using graphs

Q2. Which of the following represents the correct order of data analytics types based on workflow?
a) Predictive → Descriptive → Prescriptive → Diagnostic
b) Descriptive → Diagnostic → Predictive → Prescriptive
c) Diagnostic → Descriptive → Prescriptive → Predictive
d) Descriptive → Predictive → Diagnostic → Prescriptive

Q3. Under which level of measurement does the variable "height of a person" fall?
a) Nominal
b) Ordinal
c) Interval
d) Ratio

Q4. In Python (Pandas), which command is used to select rows by integer position?
a) df.loc[]
b) df.select()
c) df.iloc[]
d) df.index[]

Q5. Which of the following is NOT a benefit of using Jupyter Notebook?
a) Easy documentation
b) User-friendly interface
c) Faster hardware execution
d) Web-based code editing

Q6. The arithmetic mean is NOT appropriate for which type of data?
a) Interval
b) Ratio
c) Ordinal
d) Nominal

Q7. If a distribution is positively skewed, which relationship holds true?
a) Mean < Median < Mode
b) Mode < Median < Mean
c) Mean = Median = Mode
d) Median < Mean < Mode

Q8. Which measure is used to compare relative dispersion between two datasets?
a) Variance
b) Standard deviation
c) Range
d) Coefficient of variation

Q9. A dataset has a mean of 50 and a standard deviation of 10. According to the Empirical Rule, approximately what percentage of observations lie between 30 and 70?
a) 68%
b) 75%
c) 95%
d) 99.7%

Q10. Consider the following Python code:

```
"x = [10, 20, 30]
y = x
y.append(40)
print(len(x))"
```

What will be the output?
a) 3
b) 4
c) Error
d) 1

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